

Land Use Land Cover Change Detection and Future Prediction using ANN-CA Model in QGIS MOLUSCE

Project Report

Submitted By

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Certificate

This is to certify that the project entitled “**Land Use Land Cover Change Detection and Future Prediction using ANN-CA Model in QGIS MOLUSCE**” is a bonafide work carried out by:

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Prof. Uttam Kumar
(Project Guide)

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Abstract

Land Use Land Cover (LULC) analysis is an important aspect of environmental monitoring and urban planning. Rapid urbanization and changing land patterns significantly impact ecological balance, vegetation cover, and natural resources. Therefore, accurate monitoring and prediction of land use changes are essential for sustainable development and planning.

This project focuses on detecting historical land use changes and predicting future LULC patterns using the Artificial Neural Network and Cellular Automata (ANN-CA) model implemented through the MOLUSCE plugin in QGIS.

LULC maps for the years 2010, 2015, and 2020 were prepared using satellite imagery. Spatial driving factors such as NDVI, NDBI, slope, and Euclidean distance rasters from roads, railways, and water bodies were generated and aligned for modelling. The ANN model learned transition probabilities from historical LULC changes, while Cellular Automata incorporated neighborhood effects for spatial simulation.

Validation was carried out using Kappa statistics and accuracy assessment methods. Finally, the trained model was used to predict the future LULC map for the year 2030.

The results indicate continuous urban expansion, reduction in vegetation areas, and changes in barren land distribution. The study demonstrates that ANN-CA modelling in QGIS MOLUSCE is effective for future land use simulation and urban growth prediction.

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Chapter 1

Introduction

Land Use Land Cover (LULC) change analysis is one of the most important applications of Remote Sensing and Geographic Information Systems (GIS). It helps in understanding environmental transformations, urban growth, deforestation, vegetation degradation, and changes in water bodies over time.

Rapid urbanization and population growth have significantly altered natural landscapes in recent decades. Monitoring these changes is essential for sustainable urban planning, environmental conservation, and resource management.

Remote sensing provides temporal satellite imagery for detecting land use changes, while GIS enables spatial analysis and visualization. Machine learning techniques such as Artificial Neural Networks (ANN) combined with Cellular Automata (CA) have become powerful tools for modelling future land use dynamics.

Artificial Neural Networks learn complex transition relationships between land classes and spatial drivers, whereas Cellular Automata incorporate neighborhood influence and spatial dependency into the simulation process.

In this project, the MOLUSCE plugin of QGIS was used to perform:

- Historical LULC change detection
- Transition probability modelling
- ANN training and validation
- Future LULC prediction for 2030

Chapter 2

Objectives

The major objectives of this project are:

- To generate LULC maps for 2010, 2015, and 2020.
- To analyze historical land use changes.
- To prepare spatial driver variables such as NDVI, NDBI, slope, and distance rasters.
- To evaluate correlation among spatial variables.
- To train an Artificial Neural Network model using historical transitions.
- To simulate future land use patterns using Cellular Automata.
- To predict the LULC map for the year 2030.

Chapter 3

Study Area

The study area selected for this project is an urban region located in India. The area lies approximately between latitudes 12.79°N to 13.20°N and longitudes 77.29°E to 77.80°E. The region contains a mixture of built-up land, vegetation, barren land, and water bodies, making it highly suitable for Land Use Land Cover (LULC) analysis and urban growth prediction.

The study area has experienced rapid urbanization and infrastructure development over the past decade. Increasing population growth and expansion of transportation networks have significantly influenced land use transformation in the region. Monitoring these changes is essential for sustainable urban planning and environmental management.

The region exhibits noticeable transitions from vegetation and barren land into built-up areas, particularly around transportation corridors and urban centers. Such dynamic land use patterns make the area suitable for applying Artificial Neural Network and Cellular Automata (ANN-CA) based simulation models.

This project was inspired by previous research studies conducted on the Accra Region in Ghana, where ANN-CA modelling techniques were successfully applied for future LULC prediction. The methodology and workflow adopted in this project were based on concepts derived from those research works, while the present analysis was performed using datasets corresponding to the selected Indian study region.

Chapter 4

Data Used

The following datasets were used for the study.

Dataset	Source	Resolution	Purpose
Landsat Satellite Imagery	USGS / GEE	30m	LULC Classification
NDVI	Google Earth Engine	30m	Vegetation Analysis
NDBI	Google Earth Engine	30m	Built-up Analysis
DEM/Slope	SRTM	30m	Terrain Analysis
Road Network	OpenStreetMap	Vector	Distance Raster
Railway Network	OpenStreetMap	Vector	Distance Raster
Water Bodies	OpenStreetMap	Vector	Distance Raster

Table 4.1: Datasets Used in the Study

Chapter 5

Methodology

5.1 Research Inspiration

This project was inspired by the research paper titled:

“Predicting Land Use Land Cover Changes and Impact on Urban Wetlands Using Cellular Automata and Artificial Neural Networks Approach, A Case Study in Greater Accra, Ghana”

The methodology presented in the paper was adapted and implemented for an Indian urban study region using the QGIS MOLUSCE plugin and ANN-CA modelling techniques.

Several concepts such as transition probability analysis, driver variable selection, and ANN-based Cellular Automata simulation were inspired by the referenced research work.

5.2 Overall Workflow

The methodology adopted in this study is shown below.

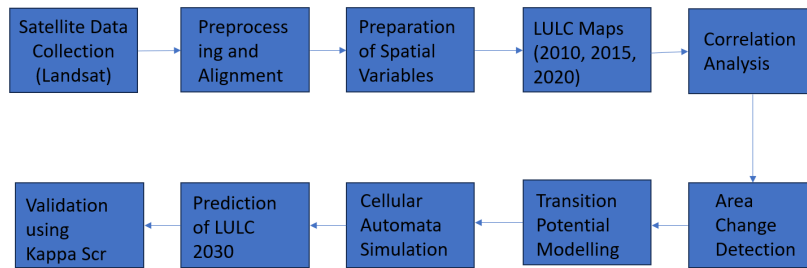


Figure 5.1: Methodology Workflow

5.3 Data Preprocessing

The collected raster datasets were preprocessed before analysis.

The preprocessing steps included:

- Clipping raster datasets to the study area
- Reprojection into common CRS
- Raster alignment
- Resampling
- Conversion into compatible formats

All raster layers were aligned to ensure identical:

- spatial resolution
- extent
- projection
- pixel dimensions

5.4 LULC Classification

The study area was classified into four major land use classes:

Class	Description
0	Waterbody
1	Built-up Area
2	Vegetation
3	Barren Land

Table 5.1: LULC Classes

5.5 Preparation of Spatial Drivers

Spatial driving variables influencing land use change were prepared.

These included:

- NDVI (Normalized Difference Vegetation Index)
- NDBI (Normalized Difference Built-up Index)
- Slope
- Euclidean Distance from Roads
- Euclidean Distance from Railways
- Euclidean Distance from Water Bodies

NDVI represents vegetation density, while NDBI highlights built-up areas. Distance rasters help model urban expansion patterns and accessibility influence.

5.5.1 Driver Variables Used in Simulation

Several spatial driver variables were used to model land use transitions and urban growth patterns. These variables influence the probability of land transformation and improve prediction accuracy.

The driver variables used in this study include:

- NDVI (Normalized Difference Vegetation Index)
- NDBI (Normalized Difference Built-up Index)
- Euclidean Distance from Roads

- Euclidean Distance from Secondary Roads
- Euclidean Distance from Railway Stations
- Euclidean Distance from Water Bodies
- Slope

These variables were selected because they significantly influence urban expansion, vegetation distribution, accessibility, and land conversion processes.

5.6 Correlation Analysis

Correlation analysis was performed in MOLUSCE using Pearson's Correlation method to identify redundancy among spatial variables.

Variables with very high correlation were avoided to improve model efficiency and reduce overfitting.

	EucDist_waterbody_aligned	Slope_aligned	NDBI_2020_new_aligned	NDVI_2020_new_aligned
EucDist_railwaySt_aligned	0.6102139810822633	0.12607800332877223	0.14955393561461436	0.06984767163095079
EucDist_roadnetwork_aligned	0.3705928516694298	0.09027960634232357	0.06604735536084727	0.14812891437248207
EucDist_secondaryRoads_aligned	0.32157737056417773	0.11770830132916955	0.0839415318240339	0.1553802675477743
EucDist_waterbody_aligned	--	0.14109442867491098	0.17311066364493144	0.09942799533127773
Slope_aligned		--	0.03206649291553899	0.00807458074564994
NDBI_2020_new_aligned			--	-0.5912262349299428
NDVI_2020_new_aligned				--

Figure 5.2: Correlation Matrix of Spatial Variables

5.7 Area Change Analysis

The area change module was used to calculate:

- land use transitions

- percentage changes
- transition matrix

A change map was generated for ANN training.

Class statistics

raster units

Class color	2015	2020	Δ	2015 %	2020 %	Δ %	
1	0.07 sq. unknown	0.08 sq. unknown	0.00 sq. unknown	37.41927174039243	39.08924010217114	1.6699683617787073	
2	0.07 sq. unknown	0.03 sq. unknown	-0.04 sq. unknown	33.59910164867062	15.496306455358178	-18.10279519331244	
3	0.04 sq. unknown	0.03 sq. unknown	-0.01 sq. unknown	19.49734413096482	16.20123650296064	-3.2961076280041794	
4	0.02 sq. unknown	0.06 sq. unknown	0.04 sq. unknown	9.484282479972135	29.21321693951004	19.728934459537903	

Transition matrix

	1	2	3	4	
1	0.4701488846979885	0.14225445660236488	0.0694662742949672	0.31813038440467933	
2	0.4843231160472763	0.25791619197771165	0.2121290002105729	0.04563169176443909	
3	0.13356519991514354	0.034525045684787994	0.03357785395010589	0.7983319004499626	
4	0.27620676199876054	0.08797178292438466	0.6136295820230909	0.022191873053763935	

Figure 5.3: Area Change Statistics and Transition Probability Matrix between 2015 and 2020

The transition matrix illustrates the probability of conversion from one land use class to another between 2015 and 2020. Significant transitions were observed from vegetation and barren land into built-up regions, indicating rapid urban expansion in the study area.

5.8 ANN Transition Potential Modelling

Artificial Neural Network (Multi-layer Perceptron) was used for transition potential modelling.

The ANN learned transition probabilities from historical LULC maps and spatial variables.

The following parameters were used:

Parameter	Value
Method	ANN Multi-layer Perceptron
Neighbourhood	3 px
Learning Rate	0.01
Maximum Iterations	300
Hidden Layers	3
Momentum	0.05

Table 5.2: ANN Parameters Used

The ANN model learned transition behavior between different land use classes.

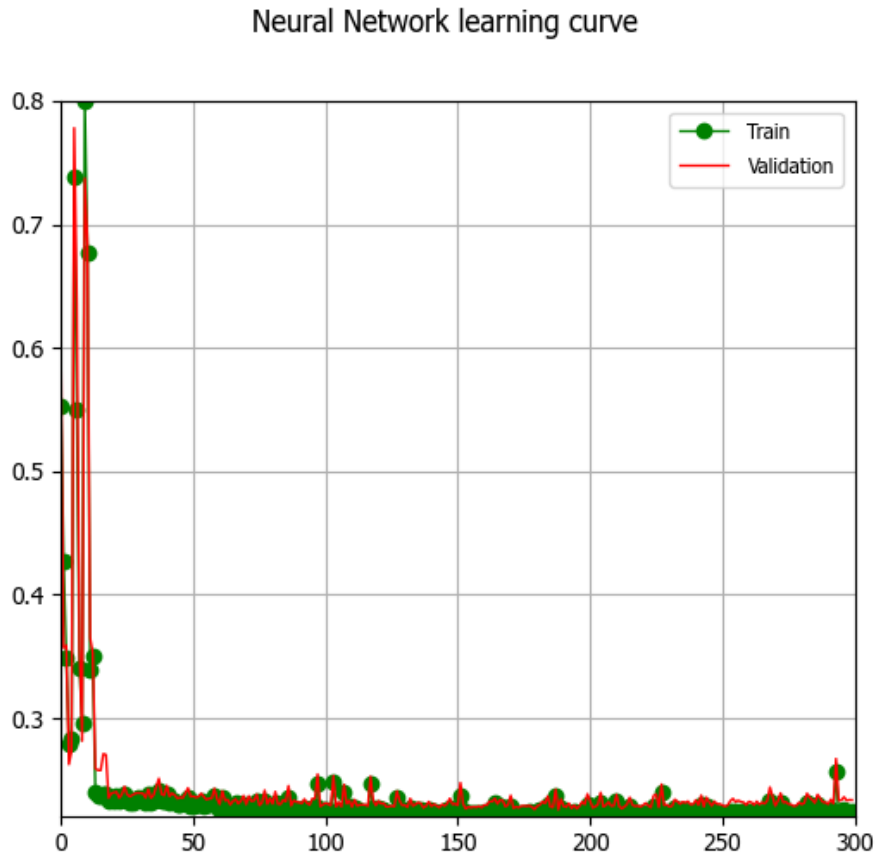


Figure 5.4: ANN Training and Validation Curve

5.9 Cellular Automata Simulation

Cellular Automata simulation was used to predict future land use changes.

The CA model considered:

- neighborhood influence
- spatial dependency

- transition probability

The trained ANN model and transition potential maps were integrated into CA simulation to predict future LULC for 2030.

Chapter 6

Results and Discussion

6.1 LULC Maps

6.1.1 LULC Map 2010

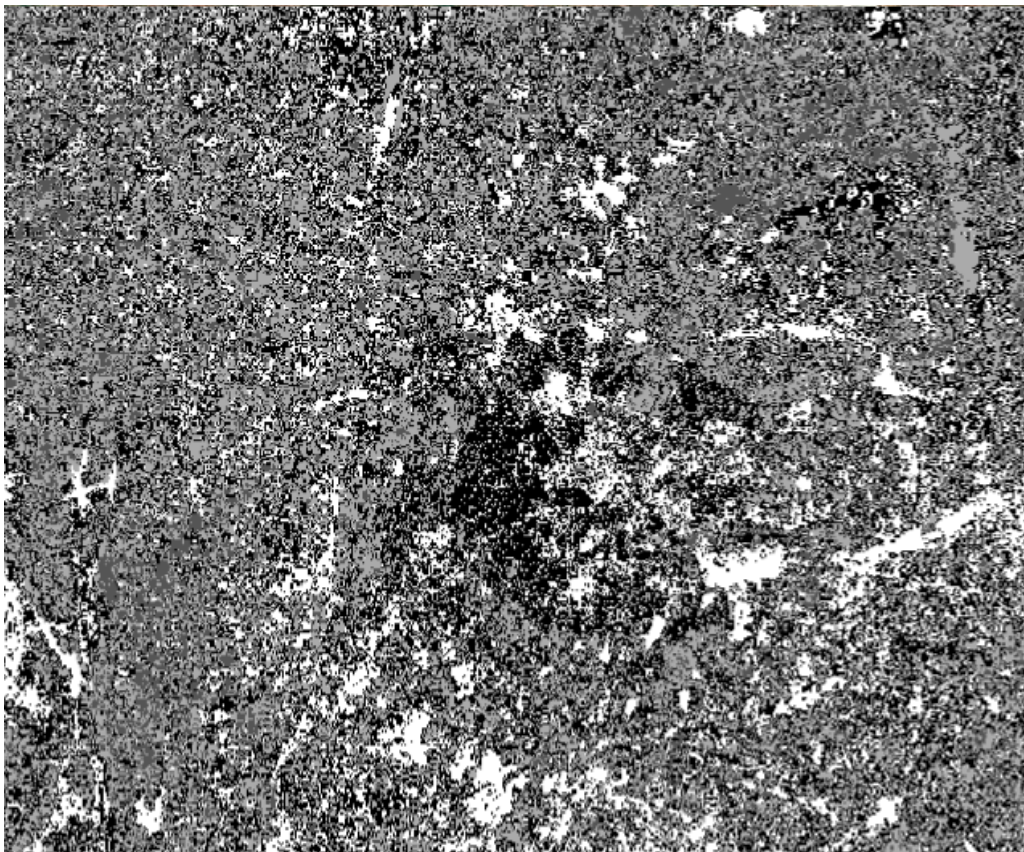


Figure 6.1: LULC Map of 2010

6.1.2 LULC Map 2015

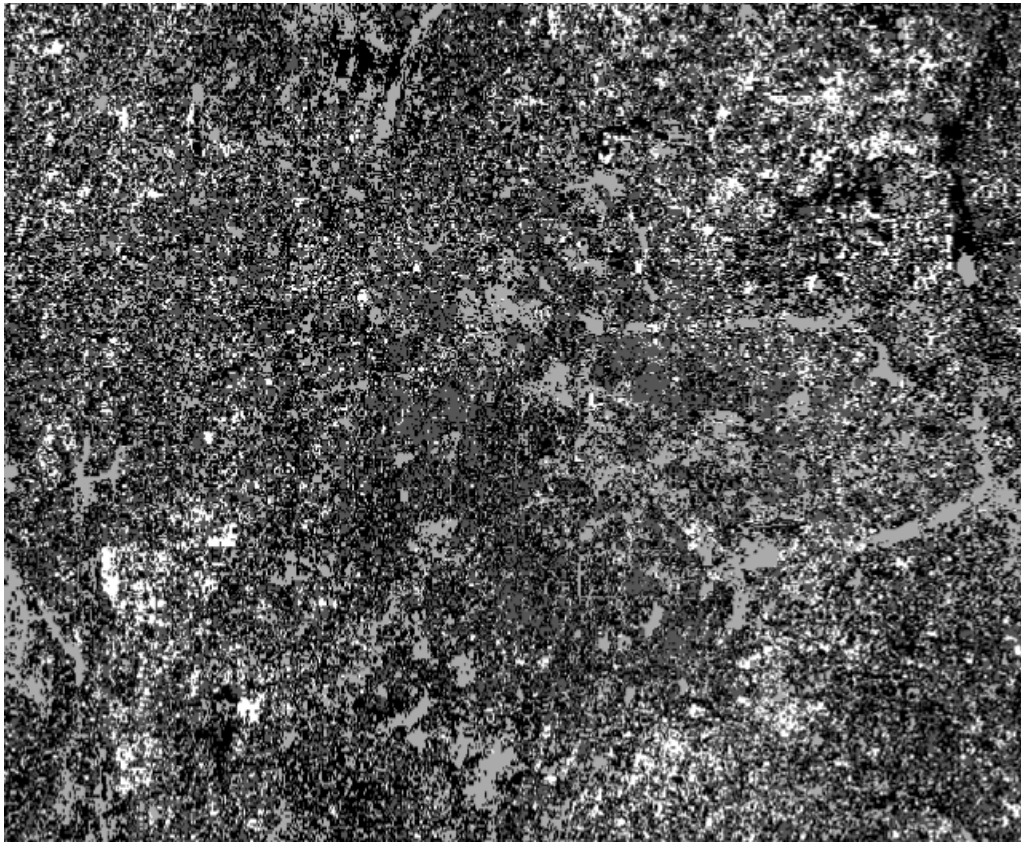


Figure 6.2: LULC Map of 2015

6.1.3 LULC Map 2020

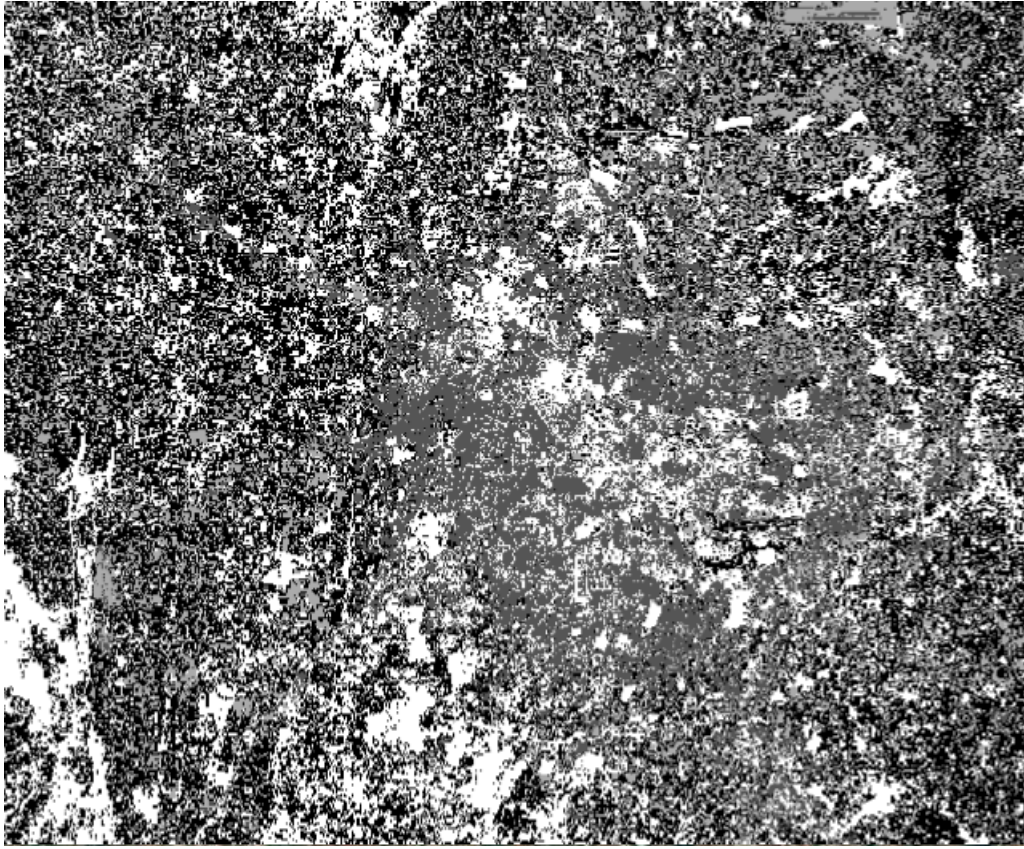


Figure 6.3: LULC Map of 2020

6.2 Historical Land Use Change Analysis

The analysis revealed noticeable urban expansion between 2010 and 2020. Vegetation areas decreased in regions experiencing rapid built-up growth. Barren land and waterbody distributions also changed during the study period.

Urban growth was primarily concentrated around transportation corridors and central urban regions.

6.3 Discussion of Urban Growth Patterns

The historical analysis revealed continuous expansion of built-up areas between 2010 and 2020. Urban growth was primarily concentrated around transportation corridors, road networks, and central urban regions.

Vegetation areas showed a declining trend due to increasing infrastructure development and urban sprawl. Barren land regions also experienced fragmentation and conversion into built-up areas.

The observed spatial patterns indicate strong influence of accessibility factors such as proximity to roads and railway networks on future urban expansion.

The ANN-CA model successfully captured these transition patterns and generated realistic spatial simulations for future land use prediction.

6.4 ANN Model Performance

The Artificial Neural Network (ANN) based transition potential model was trained using the Multi-Layer Perceptron (MLP) method available in the MOLUSCE plugin. The training process utilized spatial explanatory variables including NDVI, NDBI, slope, and Euclidean distance layers derived from road networks, railway lines, and water bodies.

The ANN learning curve demonstrated gradual convergence between the training and validation datasets, indicating stable learning behaviour and acceptable transition modelling performance. The model parameters used during training included:

- Neighbourhood size: 3 pixels
- Learning rate: 0.010
- Hidden layers: 3
- Momentum: 0.050
- Maximum iterations: 300

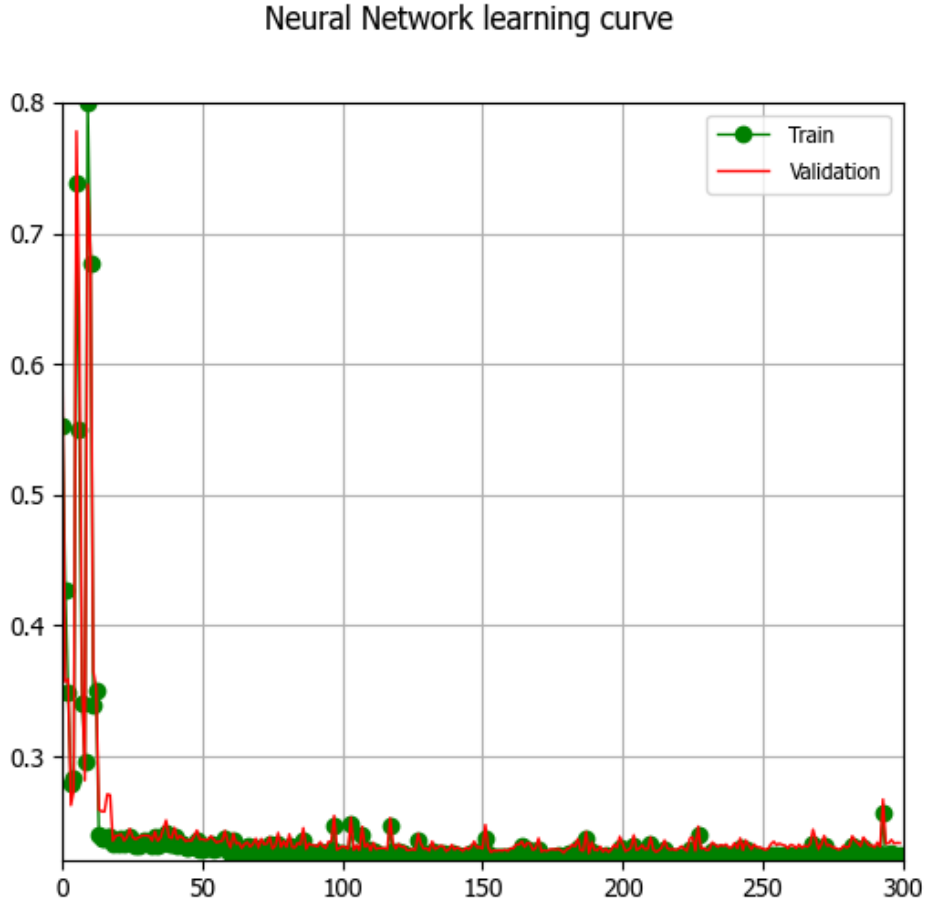


Figure 6.4: ANN Training and Validation Learning Curve

The ANN model successfully learned transition probabilities from historical land use changes between 2010 and 2015, which were later used in Cellular Automata simulation for future LULC prediction.

6.5 Validation and Accuracy Assessment

Since actual ground truth data for the year 2030 is not available, direct validation of the predicted 2030 LULC map was not possible. Therefore, model validation was performed by predicting the LULC map for the year 2020 using historical datasets from 2010 and 2015, and then comparing the predicted 2020 map with the actual classified LULC map of 2020.

The MOLUSCE validation module was used to calculate multiple validation metrics including:

- Overall Percentage Correctness

- Kappa Overall
- Kappa Histogram
- Kappa Location

The obtained validation results are summarized below:

- Percentage Correctness: 26.93%
- Kappa Overall: 0.02055
- Kappa Histogram: 0.68396
- Kappa Location: 0.03005

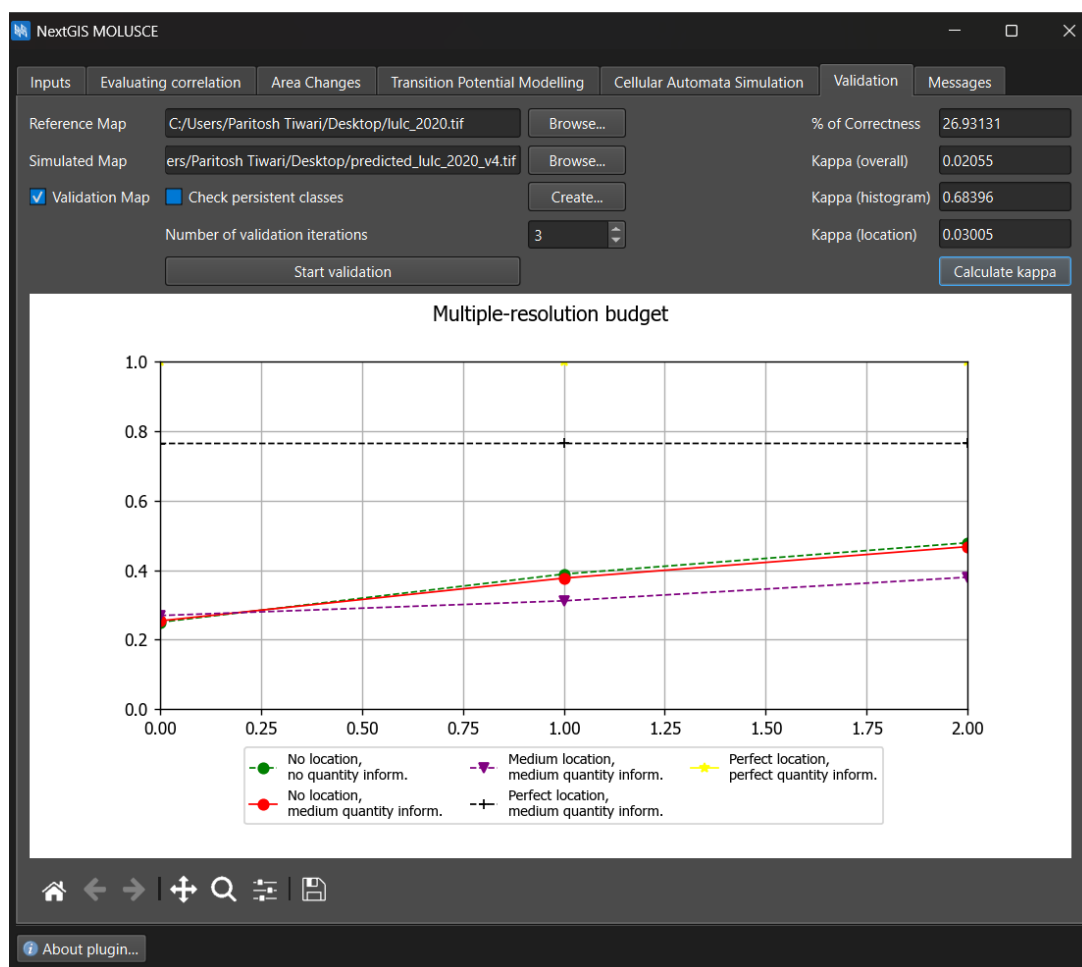


Figure 6.5: Validation Results and Multi-Resolution Budget Analysis for Predicted LULC 2020

Although the overall kappa coefficient is relatively low, the histogram kappa value indicates that the model was able to reasonably capture the quantity distribution of land

use classes. The validation results also demonstrate the challenges associated with long-term spatial prediction using limited historical temporal datasets and coarse classification boundaries.

The validated ANN-CA framework was subsequently utilized to generate the predicted LULC map for the year 2030.

6.6 Predicted LULC Map for 2030

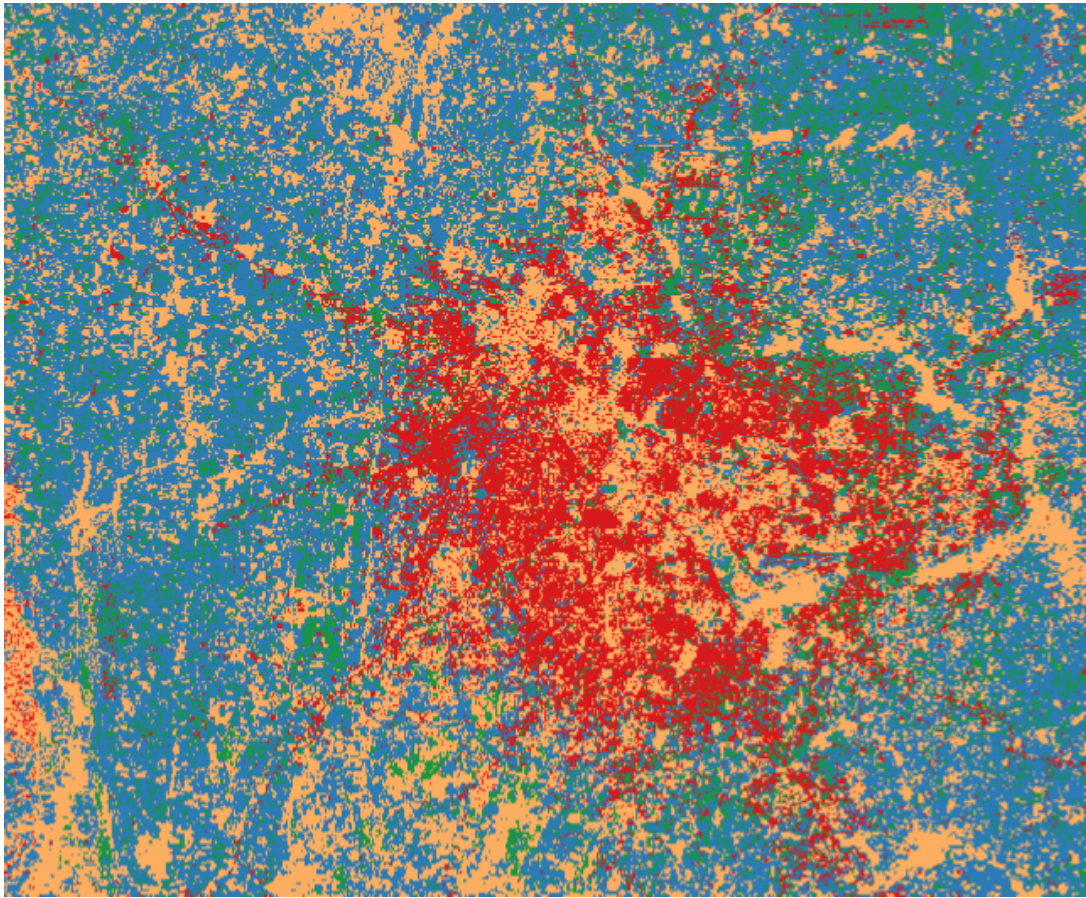


Figure 6.6: Predicted LULC Map for 2030

The predicted LULC map for 2030 indicates:

- significant urban expansion
- reduction in vegetation cover
- fragmented barren land distribution
- influence of transportation infrastructure on growth

The results suggest continued urbanization in the study area over the coming decade.

Chapter 7

Limitations of the Study

The present study has several limitations:

- Validation of the predicted 2030 LULC map was not possible due to the absence of future ground truth data.
- Medium-resolution satellite imagery may introduce classification confusion between land use classes.
- Prediction accuracy depends heavily on the quality of historical datasets and selected driver variables.
- Socioeconomic and climatic variables were not incorporated into the simulation due to limited data availability.
- ANN-CA models are sensitive to parameter selection such as learning rate, hidden layers, and neighborhood size.

Despite these limitations, the study demonstrates the effectiveness of ANN-CA modelling for future urban growth prediction and land use simulation.

Chapter 8

Conclusion

This project successfully demonstrated Land Use Land Cover change detection and future prediction using ANN-CA modelling in QGIS MOLUSCE.

Historical LULC maps for 2010, 2015, and 2020 were analyzed to identify land use transitions. Spatial driver variables such as NDVI, NDBI, slope, and distance rasters were incorporated into the ANN model.

The Artificial Neural Network effectively learned historical transition behavior, while Cellular Automata simulated neighborhood-based spatial changes.

The predicted 2030 LULC map indicates increasing urban expansion and decreasing vegetation cover. The study demonstrates that ANN-CA models are effective tools for urban growth simulation and environmental planning.

Chapter 9

Future Scope

Future improvements can include:

- use of higher resolution satellite imagery such as Sentinel-2
- incorporation of socio-economic and climatic variables
- integration of advanced deep learning based prediction models
- use of longer multi-temporal historical datasets
- field-based validation and ground truth verification
- implementation of ensemble machine learning approaches
- improved urban policy simulation and scenario-based modelling

Chapter 10

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